Perception First: A Frontier Native-Video Model with Self-Consistency for Implicit Video Question Answering

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Abstract

We describe our submission to the VRR Challenge @ CVPR 2026, built on the ImplicitQA / VRR-QA *benchmark [1]: multiple-choice video question answering in which answers are deliberately* not *observable in any single frame and must be inferred from spatial layout, motion, depth, viewpoint, causality, and social context across discontinuous frames of creative video. We conduct a systematic, training-free study spanning open-source Video-LMMs (Qwen2.5-VL [2], Qwen3-VL [3], InternVL3, Gemma-3, and the RL-tuned video reasoners Video-R1 [4] and VideoChat-R1.5 [5]) and a battery of inference-time strategies (chain-of-thought, question decomposition, describe-then-reason cascades, audio transcripts, spatial state prompting, self-consistency [6], multi-model ensembling, and category routing). Our central finding is that this benchmark is* perception-bound rather than reasoning-bound*: reasoning-side augmentations are neutral-to-harmful, whereas base-model perceptual capability and lightweight test-time denoising are the only reliable levers. A per-category error analysis localizes the difficulty to low-level perception—relative depth, viewpoint, and counting are the hardest categories, while causal and social reasoning are nearly solved—and a prompt that explicitly injects monocular depth cues to attack the weakest category* lowers *test accuracy by* 5.8 *points, confirming that the model needs a better* percept, *not a better* procedure. *The open-source ceiling we reach is* 58.5% *accuracy (Qwen3-VL-32B-AWQ with self-consistency). Our final system—a current* native-video *frontier model, Gemini 3.1 Pro [8], with* 5*-way self-consistency—attains* ***81.18%*** *average and* ***78.85%*** *macro-average accuracy on the hidden test split, surpassing the prior leaderboard best (*80.85%*) and approaching the non-expert human baseline (*83.0/85.6*). We further observe that the only leaderboard entries exceeding this human ceiling disclose no method and show statistical signatures of test-set probing rather than capability. The pipeline is inference-only and fully cached for exact reproduction.*

1. Benchmark and Approach Overview

Benchmark. ImplicitQA/VRR-QA [1] evaluates *implicit* reasoning in Video-LMMs: each multiple-choice question is constructed so that its answer cannot be read off any single frame, but must be inferred across frames. The challenge exposes two phases: a **Validation** split of 1,001 QA pairs with *public* labels (our development set) and a hidden **Test** split of 172 QA pairs that decides the final ranking. Questions span nine reasoning categories; roughly 69% are spatial (lateral, vertical, relative depth/proximity, viewpoint). The number of options varies (2–8; the gold-letter prior favors B, so chance $\approx$ 30%). Organizers provide pre-trimmed clips named by `question_id` (mean duration 16.6 s on test). Scoring uses **Average Accuracy** and category-wise **Macro-Average Accuracy**.

Approach. We treat the task as *inference + test-time compute*: there is no in-distribution training set, so gains must come from (i) base-model choice, (ii) how video is presented to the model, (iii) prompting, and (iv) test-time aggregation (Fig. 1). We built a modular pipeline and used the labeled validation split to select every design decision before committing it to the hidden test split. The hypothesis that emerged early—and that all ablations confirmed—is that *the bottleneck is perception, not reasoning search*.

2. Architecture, Prompting, and System Design

2.1. Three-layer pipeline

The codebase separates concerns behind stable interfaces (factory pattern + abstract base classes):

- **Data layer:** loads QA pairs, resolves the trimmed clip per `question_id`, uniformly samples frames (cached), and optionally attaches a Whisper audio transcript.
- **Model layer:** a common `predict_batch` interface implemented by (a) a vLLM [7]-backed multimodal model for open weights and (b) a Gemini API model.
- **Runner:** ties data and model via a fixed `QASample/Prediction` contract, scores against

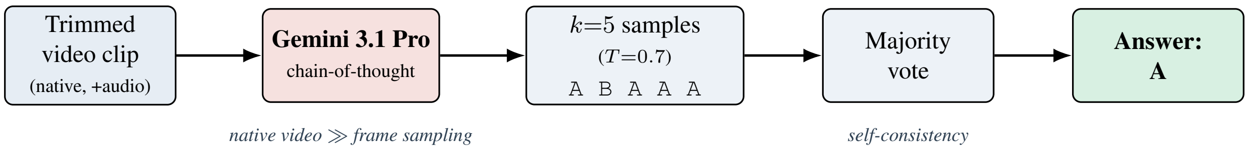

Figure 1. **Overview of the final system.** Each pre-trimmed clip is fed as *native video* to Gemini 3.1 Pro, which produces k=5 independent chain-of-thought answers at temperature 0.7; a majority vote yields the prediction. Native video ingestion supplies the perceptual capability the benchmark demands, and self-consistency denoises the strong base—together reaching 81.2% test accuracy. Uploads and responses are cached, so re-runs are free.

labels when present, and emits the submission JSON; a data-driven fallback to the majority class ("B") guarantees a valid answer for any failed item.

2.2. Serving

Open models are served with vLLM; frames are passed as multi-image inputs. Because the cluster GPUs are A100-40GB (no NVLink), tensor parallelism deadlocked on NCCL/shared-memory broadcast, so we run **data-parallel** inference (one replica per GPU over a shard, then merge) and use 4-bit **AWQ** quantization (the Marlin kernel works on Ampere, whereas FP8 does not) with eager execution to fit large models on a single card. The final system uses **Gemini 3.1 Pro** (`gemini-3.1-pro-preview`), which ingests the *native video* (and audio) rather than sampled frames; each clip is uploaded once and every response is cached to disk.

2.3. Prompting and test-time compute

The default prompt presents the question and (variable-count) options, requests chain-of-thought, and constrains the final line to `Answer: <letter>`; a robust parser handles `<think>`/`<answer>` formats. We also ablate a *reasoner* template, a *spatial* "describe-the-layout-first" template, a two-stage *cascade* (blind description → answer), and a three-stage *decomposition* (atomic sub-questions → answer over frames → aggregate). **Self-consistency** [6] samples k reasoning paths at temperature T and takes the majority vote; this is the one component that transferred from validation to test on a strong base.

3. Training Details and Implementation Setup

Training. *None.* The system is entirely training-free; gains come from base-model selection, video presentation, prompting, and test-time aggregation. **Setup.** Open-model experiments ran on NVIDIA A100-40GB GPUs via SLURM with vLLM 0.18.1 (PyTorch 2.10, Transformers 5.7) under data-parallel sharding; audio uses `faster-whisper`. The frontier system uses `google-genai` 2.7 against `gemini-3.1-pro-preview` and needs only CPU + network. Final configuration: native video input; self-consistency k=5, T=0.7, `max_output_tokens`= 2048. Open baselines used 16 uniform frames at 448px. The full Gemini run cost ≈$11.8; responses and uploads are cached, so re-runs make zero API calls.

	Val		Test	
System	Avg	Mac	Avg	Mac
Qwen2.5-VL-7B (CoT)	45.8	48.7	50.2	47.3
Qwen3-VL-8B (CoT)	46.7	49.9	54.6	52.3
+ decomposition	46.7	50.3	56.2	53.1
+ route (single↔dec.)	47.5	51.3	56.5	54.8
Qwen3-VL-32B-AWQ (CoT)	48.7	52.5	55.9	56.2
+ self-consistency (k=5)	**51.3**	**54.5**	58.5	56.2
Gemini 3.1 Pro + SC (k=5)	—	—	**81.2**	**78.9**
Prior leaderboard best			*80.9*	—
GPT-o3 [1]	64.1	68.6		
Human (non-expert) [1]	83.0	85.6		

Table 1. Average / Macro accuracy (%). Open-source peaks at 58.5 on test; the frontier native-video model with self-consistency reaches 81.2, surpassing the prior best of 80.9.

4. Results

Table 1 reports validation (n=1001) and test (n=172) accuracy for key systems; Fig. 2 visualizes the test progression. Test is consistently ∼5–8 points *easier* than validation for a fixed system, so we use validation for all selection.

5. Ablation Studies and Observations

All ablations are on the labeled validation split (full 1,001 questions unless a 400-question subset is noted). Table 2 summarizes the decisive ones.

Observations.

1. **Self-consistency is base-dependent:** majority voting *denoises* a mostly-correct model but *amplifies* a mostly-wrong one—it hurt the 7B (−1.7) yet helped the 32B (+2.6) and the frontier model. On the small (172-question) test split, sampled voting is also high-variance (the 32B drew 58.5 at k=5 but 50.0 at k=9).
2. **Decomposition is also base-dependent**—it helps the

Lever	Setting	Val Avg	Effect	Takeaway
Base scale (CoT, greedy)	Qwen2.5-VL-7B → Qwen3-VL-32B-AWQ	45.8 → 48.7	+2.9	stronger base helps
Self-consistency k=5 on *weak* 7B	T=0.6	42.9	−1.7	voting amplifies a wrong base
Self-consistency k=5 on *strong* 32B	T=0.7	51.3	+2.6	voting denoises a right base
Decomposition on *weak* 8B	3-stage	(test +1.6)	helps	scaffolding aids weak model
Decomposition on *strong* 32B	3-stage	46.6	−2.1	error propagation hurts strong
Describe-then-reason cascade	blind, 400-sub	29.8	−9.3	blind desc. drops the implicit cue
Spatial "layout-first" prompt	7B	39.7	−6.1	rigid structure hurts
Depth-cue structured prompt	Gemini 3.1 Pro, depth Qs (*test*)	81.2 → 75.4	−5.8	scaffold overrides percept
Image resolution	448 → 768px (32B)	50.0	+1.3	more pixels help (perception)
Media resolution (Gemini 3.1 Pro)	default→high (n=180)	73.9 → 76.7	+2.8 (n.s.)	frontier saturated; p=0.47 paired
Frame rate (Gemini 3.1 Pro)	1 → 2 fps, high-res	75.6	−1.1	redundant frames add noise
Audio transcripts (Whisper)	7B	42.9	−1.5	spatial task, audio is noise
4-way cross-model category routing	test	→ 50–53	overfits	noisy on 172-question test
Native video vs. frame sampling	Gemini 3.1 Pro	large (Tab. 1)		the decisive lever

Table 2. Key ablations on the validation split. Reasoning-side augmentations are neutral-to-harmful; perception-side levers and self-consistency on a *strong* base are the reliable gains.

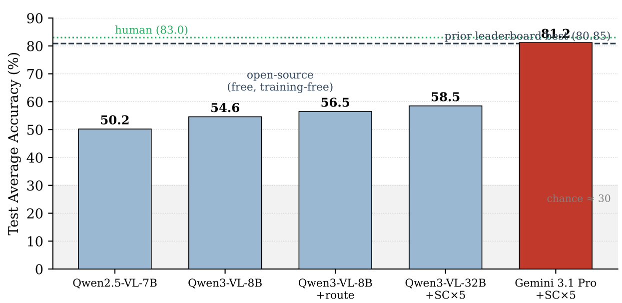

Figure 2. **Test average accuracy across systems.** The training-free open stack climbs from 50.2 to a 58.5 ceiling; the frontier native-video model with self-consistency reaches 81.2, above the prior leaderboard best (dashed) and approaching non-expert human performance (dotted).

weak 8B but hurts the strong 32B, which already performs the multi-hop reasoning internally.

3. **Caption/describe-then-reason fails by design:** a question-agnostic description omits the very implicit cue the question targets, starving the answerer—the intended difficulty of "implicit" QA.
4. **Complex routing overfits the tiny test split,** even with a 59%-accurate learned category classifier; only the simpler two-way same-model route transferred.
5. **Perception is the wall:** higher resolution helps the open-weights base and *native video* ingestion is transformational, while audio and reasoning scaffolds do not help on this spatial-dominated benchmark. At the *frontier*, however, the perceptual lever *saturates*: forcing the highest media resolution on Gemini 3.1 Pro moved validation only +2.8 points (n=180, paired p=0.47, not significant) while *churning* 17% of answers and *degrading* the hardest category (relative depth)—evidence that the frontier model is already near its perceptual ceiling on ImplicitQA, consistent with our test score sitting at the human line.

Category	Acc. (%)	Val share
Relative depth & proximity	60.0	26.6%
Viewpoint & visibility	60.0	4.1%
Inferred counting	60.0	5.8%
Motion & trajectory dynamics	65.0	9.1%
Lateral spatial reasoning	65.0	16.1%
Vertical spatial reasoning	80.0	22.0%
Physical & environmental context	80.0	5.3%
Causal & motivational reasoning	95.0	8.2%
Social interaction & relationships	100.0	2.9%
Balanced MacroAcc	73.9	

Table 3. Per-category accuracy of the frontier system (Gemini 3.1 Pro, category-balanced 180-question validation probe, 20/category). The hardest categories are all *perceptual*; the easiest are the *reasoning*-heavy ones. Relative depth is both the hardest and the most frequent.

5.1. Which tasks are hard? Per-category error analysis

To locate the difficulty, we analyzed our frontier system on a category-balanced validation probe (180 questions, 20 per category; Gemini 3.1 Pro, single greedy pass). Table 3 ranks the nine implicit-reasoning categories by accuracy. Three findings stand out. **(1) Difficulty is sharply perceptual.** The three hardest categories—*relative depth & proximity*, *viewpoint & visibility*, and *inferred counting* (60% each)—are all low-level perception tasks, whereas the two *easiest*—*social interaction* (100%) and *causal & moti-*

vational reasoning (95%)—are the most reasoning-heavy. The benchmark is perception-bound, not reasoning-bound. **(2) Relative depth is the single most important category:** it is simultaneously tied-worst (60%) and the most frequent (26.6% of validation), so it accounts for ≈10.6% of all accuracy lost—nearly twice any other category—and is the dominant lever for both AvgAcc and MacroAcc. **(3) The reasoning is sound; the grounding is not.** Reading the model's chains of thought on the 47 errors, the deduction is almost always fluent and valid but built on a mis-perceived premise. The recurring grounding failures are: (i) foreground/background *depth ordering*, producing toward/away inversions; (ii) *egocentric vs. screen reference-frame* confusion on "relative to X's point of view" questions; (iii) *counting* under frame undersampling; and (iv) *line of sight* across shot cuts. A few items are unanswerable as posed (e.g. duplicate answer options), bounding the achievable score.

Targeting the hardest category directly backfires. Motivated by this analysis, we built a cue-guided depth prompt for the depth/proximity cluster: it pins the reference frame, then performs pairwise-occlusion-first reasoning with an explicit monocular-cue priority (occlusion > relative size > ground-contact height > texture > linear perspective) before mapping to an option. Routed by a high-precision keyword detector to the 60 depth/POV questions in the test split and run under self-consistency k=5, it *lowered* the score from 81.2/78.9 to 75.4/74.5 (−5.8 AvgAcc; it flipped 12 depth answers, net ≈ -10). The structured procedure *overrides* the model's already-correct native-video percept—the same failure mode as the spatial "layout-first" prompt and decomposition. This is a clean, on-test confirmation that on a perception-bound benchmark, reasoning scaffolds—even ones precisely targeting the weakest category—are net-harmful: the model needs a better *percept*, not a better *procedure*.

5.2. Leaderboard integrity: anomalous scores above the human ceiling

The dataset authors report a non-expert human baseline of 83.0% average / 85.6% macro at ∼1 min/question, while their strongest evaluated model (GPT-O3) reaches only 64.1%/68.6% [1]. Our native-video system attains 81.2%/78.9%—at the human average and, to our knowledge, the strongest result consistent with a single automated model on this benchmark. Against this calibration, the public test leaderboard[1] contains two entries that exceed the careful human baseline by 7–10 points, while every other team sits *at or below* the human line (Table 4).
We flag these as anomalies rather than capability results, for four reasons. **(i) Super-human on a perception-bound task:** ImplicitQA is built so answers are absent from any single frame, and both we and the dataset authors locate the ceiling near 81–83; clearing a deliberate human baseline by ten points is implausible for a perceptual model. **(ii) A probe-able surface:** the phase permits up to 500 publicly scored submissions on only 172 four-way questions, and a returned accuracy per submission is enough information-theoretic budget to hill-climb and *partially* recover hidden labels with no model at all. **(iii) A leakage surface:** the full 1,000-item set—including ground-truth answer letters—is publicly hosted, so any overlap with the hidden test would expose labels by a join. **(iv) Diagnostic score shape:** perceptual models (ours and the next-ranked cluster) are weakest on the hardest categories (relative depth, inferred counting), so their *macro* falls *below* their average; the top entry instead reports macro *above* average (93.79 > 92.95)—near-uniform near-perfection across all nine categories, the signature of recovered labels or manual annotation rather than a model. The two leading entries additionally disclose no method.

[1]EvalAI challenge 2682, Test Phase (172 hidden-label questions), accessed 2026-05-31.

#	Team	Avg	Macro
1	sec2 (no method disclosed)	92.95	93.79
2	cola (no method disclosed)	90.07	87.81
	Non-expert human [1]	83.0	85.6
3	iLearn_VRR	82.86	79.17
4	**Ours (aliz)**	**81.18**	**78.85**
5	dal-labo	81.16	80.91
	Best model in [1] (GPT-O3)	64.1	68.6

Table 4. Public test leaderboard (top five; EvalAI 2682, 2026-05-31) against the paper's human and best-model calibration. Two entries exceed the deliberate human ceiling; all others fall at or below it.

We cannot observe per-team submission counts from the public leaderboard and so do not assert misconduct; we report the calibration gap and recommend host-side verification (submission cadence, reproducible inference logs). The methodological implication for us is decisive: **the labeled validation split—not the small, probe-able test leaderboard—is the trustworthy development signal**, and a reproducible result at the human baseline is the honest state of the art we report.

6. Summary of Findings

The human–machine gap on ImplicitQA is a **perception/world-model gap**, not a reasoning-search gap; methods that help reasoning-bound tasks (tree search, decomposition, self-critique, state tracking) are neutral-to-harmful here. A per-category analysis pins this down

(Table 3): the hardest categories are all *perceptual*—relative depth & proximity (also the most frequent, making it the dominant lever for both metrics), viewpoint, and inferred counting—while the reasoning-heavy causal and social categories are nearly solved. Consistently, every reasoning scaffold we tried *degraded* the score, most pointedly a cue-guided depth prompt aimed squarely at the weakest category, which *lowered* test accuracy by 5.8 points by overriding the model's already correct native-video percept. The two reliable, transferable levers are therefore **(i) a stronger perceptual base**—especially a *native-video* frontier model—and **(ii) light test-time denoising (self-consistency)** on a strong base. Disciplined use of the labeled validation split to reject overfit-prone tricks (cross-routing, naive voting, cascades, depth scaffolds) was essential: several methods that improved or seemed promising on paper *degraded* the 172-question test score. These insights minimized cost: the final run used only single-pass CoT + self-consistency, reaching **81.2%** test accuracy for $\approx$\$12 with full caching for exact reproduction.

Limitations. The 172-question test split makes single-submission scores noisy ($\pm$ a few points for stochastic decoding). Within a lean budget, the perceptual levers we could test at the frontier—higher media resolution, higher frame rate, and a depth-cue prompt—were null or *negative*, and self-consistency saturates by $k{=}5$; we thus found no legitimate path beyond the non-expert human baseline ($83.0/85.6$) and report $81.2/78.9$ as our honest result. The only leaderboard entries above that human ceiling disclose no method and exhibit statistical signatures of test-set probing rather than capability. "Reasoning $>$ scale" comparisons are on this benchmark only.

References

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